

CHINO, CA, USA

WMO: 722899

Lat: 33.975N

Lon: 117.636W

Elev: 198

StdP: 98.97

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03179

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.3	2.6	-14.3	1.1	18.5	-11.0	1.5	18.9	10.9	17.0	9.2	17.4	0.8	10	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.6	38.4	20.9	36.8	20.6	35.0	20.0	23.1	34.6	22.3	33.4	21.5	32.3	5.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.7	27.1	18.5	13.6	26.2	17.6	12.9	25.1	69.1	34.8	65.8	33.5	63.0	32.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.7	7.1	-1.4	42.3	1.4	2.4	-2.4	44.0	-3.2	45.4	-4.0	46.7	-5.0	48.5
			-2.6	24.5	1.3	1.6	-3.5	25.6	-4.3	26.6	-5.0	27.5	-6.0	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	12.8	12.8	14.7	16.4	19.1	22.0	25.2	25.4	23.9	19.9	15.6	12.2
	DBStd	5.60	3.18	3.17	3.18	3.27	2.82	2.78	2.52	2.43	3.17	3.15	3.22	2.89
	HDD10.0	33	9	7	2	1	0	0	0	0	0	0	2	12
	HDD18.3	856	174	156	119	76	24	3	0	0	1	19	93	191
	CDD10.0	3087	96	87	149	193	281	361	472	478	417	306	169	79
	CDD18.3	868	2	2	7	17	47	113	213	220	168	67	12	1
	CDH23.3	10871	94	114	220	397	658	1291	2374	2502	1914	944	306	57
	CDH26.7	5646	14	29	68	156	287	636	1333	1457	1091	458	109	7
(14) Wind	WSAvg	2.4	1.9	2.0	2.3	2.8	3.1	3.0	2.9	2.7	2.3	2.1	1.8	1.7
(15) Precipitation	PrecAvg	292	58	64	35	26	6	1	3	1	5	13	30	48
	PrecMax	621	214	214	86	74	22	2	22	5	13	140	56	216
	PrecMin	112	10	5	8	7	1	0	0	0	2	0	10	4
	PrecStd	137	57	54	21	15	5	0	5	1	3	31	13	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	29.8	31.9	34.7	36.3	39.0	40.1	40.6	40.9	37.2	33.0	27.6
		MCWB	12.8	13.7	16.0	15.8	18.3	20.2	22.1	21.2	21.2	18.1	15.1	13.2
	2%	DB	26.0	27.1	28.5	30.9	32.8	35.2	37.3	37.8	37.8	34.4	29.9	24.9
		MCWB	12.6	13.3	14.9	15.5	17.7	20.1	21.7	21.2	20.9	16.9	14.5	12.1
	5%	DB	23.7	23.8	25.8	27.8	30.0	32.7	35.3	36.1	35.7	32.0	27.0	22.5
		MCWB	11.9	12.4	14.3	15.0	17.4	19.6	21.3	20.8	20.2	16.4	13.7	11.1
	10%	DB	21.2	20.7	22.7	25.0	27.1	30.5	33.4	34.0	33.0	28.8	23.9	19.7
		MCWB	11.0	11.8	13.7	14.5	17.0	19.0	20.9	20.5	19.4	16.0	12.9	10.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	16.2	18.4	19.0	21.5	22.4	24.8	23.9	23.7	20.9	17.2	15.7
		MCDB	22.1	22.7	27.5	29.9	32.1	33.9	34.6	36.3	36.2	30.8	24.9	20.6
	2%	WB	14.6	14.8	16.6	17.2	19.7	21.1	23.3	22.7	22.6	19.5	16.0	14.4
		MCDB	20.4	22.1	25.1	26.9	30.2	32.3	34.0	34.6	34.5	28.9	25.0	19.1
	5%	WB	13.6	13.8	15.4	16.0	18.4	20.2	22.3	21.8	21.4	18.4	15.2	13.4
		MCDB	19.4	20.9	23.2	25.3	27.8	30.9	32.7	33.4	32.6	27.6	23.2	18.2
	10%	WB	12.6	12.8	14.4	15.0	17.2	19.4	21.4	21.1	20.4	17.4	14.3	12.3
		MCDB	18.2	18.6	20.8	23.2	25.6	29.4	31.5	32.2	30.6	25.3	21.4	17.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	13.9	14.3	14.8	14.4	15.6	16.8	17.6	17.4	16.3	16.2	14.8
		MCDBR	19.9	20.7	20.1	20.9	20.6	19.9	18.8	20.1	21.3	22.3	21.4	20.1
		MCWBR	9.1	9.3	8.9	8.3	8.0	7.0	6.2	6.6	6.9	7.9	8.7	9.5
	5% WB	MCDBR	15.2	16.3	16.7	17.7	17.8	18.2	17.2	18.0	18.5	17.8	17.2	14.8
		MCWBR	7.9	8.1	8.1	7.9	7.6	6.9	6.4	6.4	6.8	7.2	7.9	7.8
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.341	0.354	0.355	0.366	0.379	0.363	0.348	0.343	0.304	0.303
	taud		2.543	2.533	2.443	2.412	2.442	2.420	2.448	2.480	2.509	2.498	2.596	2.594
	Ebn at Noon		908	936	935	932	929	913	900	912	915	894	909	897
	Edh at Noon		81	91	108	116	114	117	113	107	100	93	76	73
(44) All-Sky Solar Radiation	RadAvg		3.22	4.11	5.53	6.72	7.34	7.86	7.62	7.22	6.14	4.67	3.58	2.92
	RadStd		0.22	0.41	0.28	0.31	0.41	0.41	0.36	0.19	0.19	0.29	0.18	0.21

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (51 neighbors)	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		+1.48	+0.69	-3.27	+1.14	N/A	N/A	N/A	-230	+552
		+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223

Nomenclature: See separate page